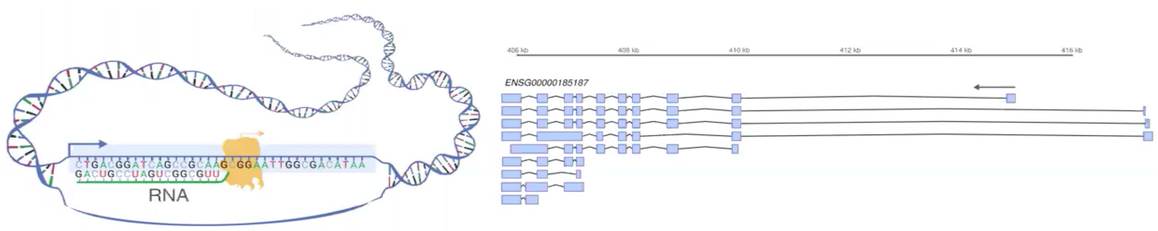
Integrating long read RNA-seq data into Bioconductor workflows with NanoporeRNASeq

Yuk Kei Wan[1], Ying Chen[1], Jonathan Göke[1]
[1]Genome Institute of Singapore, A*STAR, Singapore

Abstract

The NanoporeRNASeq package is the first long-read RNA-seq ExperimentData package available on Bioconductor. The data was generated from Oxford Nanopore Sequencing and consists of six samples from two human cell lines, namely K562 and MCF7, each with three replicates (one from direct RNA sequencing and two from direct cDNA sequencing). The six samples in this package are aligned to chromosome 22 (Grch38) in bam file format and are a subset of samples from the Singapore Nanopore Expression Consortium (SG-NEx), which provides comprehensively benchmarked datasets from three Nanopore long-read RNA-seq protocols (direct RNA, direct cDNA, and PCR cDNA) being run on five cancer cell lines (A549, Hct116, HepG2, K562, and MCF7). The NanoporeRNASeq package can be used for illustrating and evaluating Bioconductor workflows for transcript discovery, quantification, and differential expression analysis of long-read Nanopore RNA-seq.

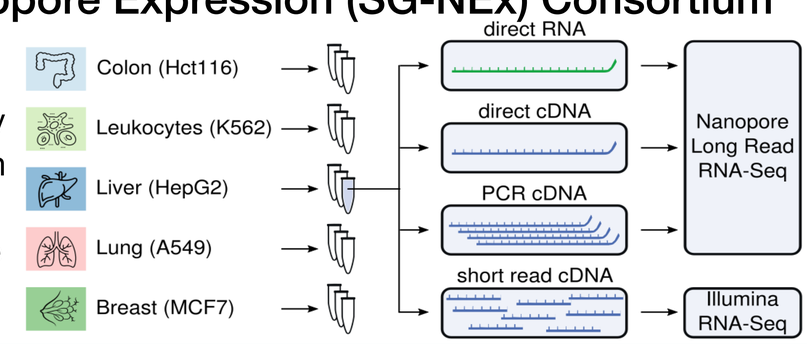
Why use Nanopore RNA-seq?



- Nanopore RNA-seq provides full-length transcript isoform reads, allowing investigations into complex post-transcriptional events.

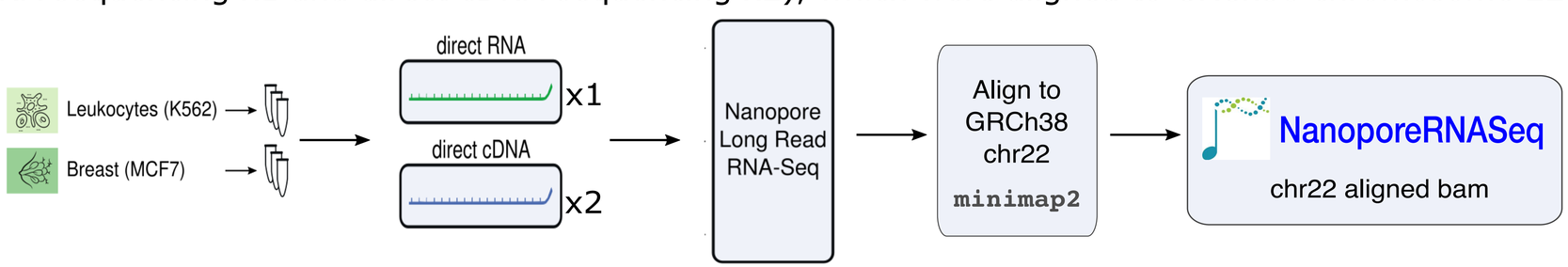
The Singapore Nanopore Expression (SG-NEx) Consortium

- The SG-NEx project, provides comprehensively benchmarked datasets on five cancer cell lines and three long-read Nanopore RNA-seq protocols.

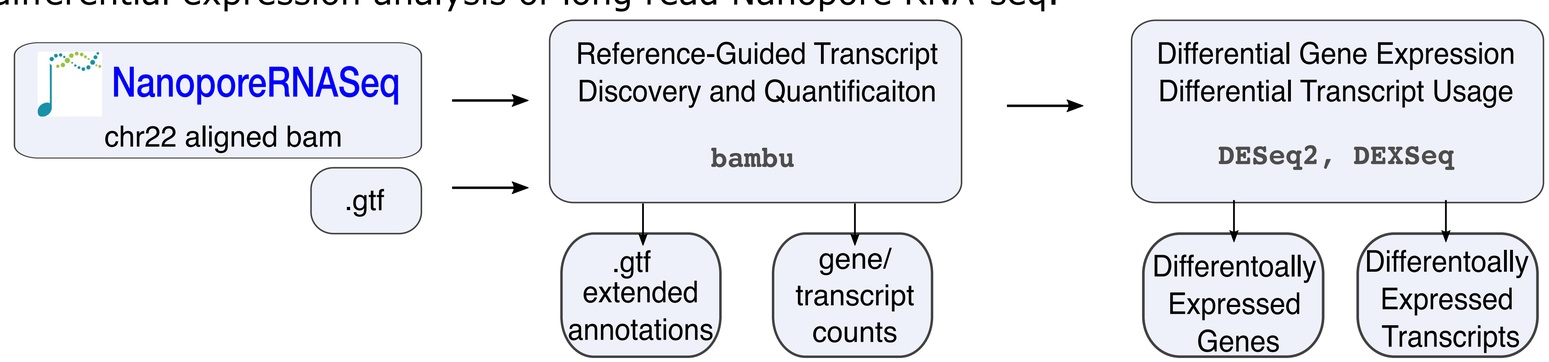


Resource: NanoporeRNASeq illustrates the use of Bioconductor workflows on long read Nanopore RNA-seq data analyses.

-NanoporeRNASeq consists of six samples from K562 and MCF7, each with three replicates (direct RNA sequencing x1 and direct cDNA sequencing x2), which were aligned to GRCh38 chromosome 22.



-NanoporeRNASeq illustrates the use of Bioconductor workflows for transcript discovery, quantification and differential expression analysis of long read Nanopore RNA-seq.



Conclusion

- We developed NanoporeRNASeq, the first long read RNA Experiment package on Bioconductor to illustrate and evaluate the use of Bioconductor packages for Nanopore RNA-seq data analyses.
- NanoporeRNASeq includes six subsetted SG-NEx samples from two human cell lines sequenced with Nanopore's direct RNA and direct cDNA protocols.

References

1. Chen, et al. (2021). A systematic benchmark of Nanopore long read RNA sequencing for transcript level analysis in human cell lines. bioRxiv.
2. Chen, et al. (2020). *Reference-guided transcript discovery and quantification for long read RNA-Seq data.* ISMB Virtual Conference

Acknowledgements

We would like to thank the Agency for Science, Technology and Research (A*STAR), Singapore for funding the SGNEx project and all contributors of the SGNEx project.